

TASK 1:DATA CLEANING AND PREPROCESSING

Import Required Libraries

```
In [3]: import pandas as pd
```

Load the Dataset

```
In [4]: df = pd.read_csv(r"C:\Users\Admin\Downloads\Sentiment dataset.csv")
```

Check or inspect data

```
In [5]: df.head()
```

Out[5]:

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023
2	2	2	Just finished an amazing workout! 💪 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023
3	3	3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023
4	4	4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023



Check Data Structure

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 732 entries, 0 to 731
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0.1    732 non-null   int64
1   Unnamed: 0      732 non-null   int64
2   Text            732 non-null   object
3   Sentiment       732 non-null   object
4   Timestamp       732 non-null   object
5   User            732 non-null   object
6   Platform        732 non-null   object
7   Hashtags        732 non-null   object
8   Retweets        732 non-null   float64
9   Likes           732 non-null   float64
10  Country         732 non-null   object
11  Year            732 non-null   int64
12  Month           732 non-null   int64
13  Day             732 non-null   int64
14  Hour            732 non-null   int64
dtypes: float64(2), int64(6), object(7)
memory usage: 85.9+ KB
```

Checking for Missing Value - No missing value were found in the dataset

```
In [7]: df.isnull().sum()
```

```
Out[7]: Unnamed: 0.1    0
        Unnamed: 0      0
        Text          0
        Sentiment      0
        Timestamp      0
        User           0
        Platform       0
        Hashtags       0
        Retweets       0
        Likes          0
        Country        0
        Year           0
        Month          0
        Day            0
        Hour           0
        dtype: int64
```

Handling missing value for numerical values

```
In [8]: df["Likes"] = df["Likes"].fillna(df["Likes"].mean())
```

Handling missing value for text values

```
In [9]: df["Country"] = df["Country"].fillna("Unknown")
```

Check for Duplicate Value-No duplicate value found

```
In [10]: df.duplicated().sum()
```

```
Out[10]: np.int64(0)
```

```
In [11]: duplicates = df.duplicated().sum()

         print(duplicates)
```

0

Remove duplicate (In case there is duplicate values in the dataset)

```
In [12]: df = df.drop_duplicates()
```

Remove Unnecessary Columns- The dataset contain an extran column: Unnamed: 0, Unnamed: 0.1

```
In [13]: df = df.drop(columns=["Unnamed: 0", "Unnamed: 0.1"])
```

Check if the extra column is removed

```
In [14]: df.head()
```

Out[14]:

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	Hour
0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	12
1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	8
2	Just finished an amazing workout! 🏋️ ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	15
3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	18
4	Trying out a new recipe for dinner tonight ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023	1	15	19

Standardize Date Format

Convert the Timestamp column to proper datetime format

```
In [15]: df["Timestamp"] = pd.to_datetime(df["Timestamp"])
```

Check if the timestamp daterime has been formatted

```
In [16]: df["Timestamp"].head()
```

```
Out[16]: 0    2023-01-15 12:30:00  
1    2023-01-15 08:45:00  
2    2023-01-15 15:45:00  
3    2023-01-15 18:20:00  
4    2023-01-15 19:55:00  
Name: Timestamp, dtype: datetime64[ns]
```

Standardize Text Columns

Some values contain extra spaces.

Remove leading and trailing spaces:

```
In [17]: df["Sentiment"] = df["Sentiment"].str.strip()  
df["Platform"] = df["Platform"].str.strip()  
df["Country"] = df["Country"].str.strip()  
df["User"] = df["User"].str.strip()  
df["Text"] = df["Text"].str.strip()
```

check if space are removed

```
In [19]: df.head()
```

Out[19]:	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	Hour
0	Enjoying a beautiful day at the park!	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	12
1	Traffic was terrible this morning.	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	8
2	Just finished an amazing workout! 💪	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	15
3	Excited about the upcoming weekend getaway!	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	18
4	Trying out a new recipe for dinner tonight.	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023	1	15	19

Standardize Capitalization

Convert text to title case:

```
In [20]: df["Country"] = df["Country"].str.upper()  
df["Platform"] = df["Platform"].str.title()  
df["Sentiment"] = df["Sentiment"].str.title()
```

```
In [46]: df.head()
```

Out[46]:

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	Hour
0	Enjoying a beautiful day at the park!	Positive	2023-01-15 12:30:00	User 123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	12
1	Traffic was terrible this morning.	Negative	2023-01-15 08:45:00	Commuter X	Twitter	#Traffic #Morning	5.0	10.0	CANADA	2023	1	15	8
2	Just finished an amazing workout! 💪	Positive	2023-01-15 15:45:00	Fitness Fan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	15
3	Excited about the upcoming weekend getaway!	Positive	2023-01-15 18:20:00	Adventure X	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	18
4	Trying out a new recipe for dinner tonight.	Neutral	2023-01-15 19:55:00	Chef Cook	Instagram	#Cooking #Food	12.0	25.0	AUSTRALIA	2023	1	15	19

Check for Datatype

In [22]: `df.dtypes`

```
Out[22]: Text          object
Sentiment            object
Timestamp    datetime64[ns]
User            object
Platform        object
Hashtags        object
Retweets        float64
Likes           float64
Country         object
Year            int64
Month           int64
Day             int64
Hour            int64
dtype: object
```

Seperate the text in user column

```
In [38]: import re
```

```
# Insert space between lowercase and uppercase letters
# Insert space between letters and numbers
# Insert space between numbers and letters
```

```
In [44]: def clean_text(text):
text = re.sub(r'([a-z])([A-Z])', r'\1 \2', text)
text = re.sub(r'([A-Za-z])([0-9])', r'\1 \2', text)
text = re.sub(r'([0-9])([A-Za-z])', r'\1 \2', text)

return text
```

```
In [45]: df['User'] = df['User'].apply(clean_text)
```

Save the Cleaned Dataset

```
In [47]: df.to_csv("cleaned_sentiment_dataset2.csv", index=False)
```

TASK 2: Exploratory Data Analysis (EDA)

Understand the Dataset

Check Shape

```
In [24]: df.shape
```

```
Out[24]: (732, 13)
```

View Column names

```
In [25]: df.columns
```

```
Out[25]: Index(['Text', 'Sentiment', 'Timestamp', 'User', 'Platform', 'Hashtags',  
               'Retweets', 'Likes', 'Country', 'Year', 'Month', 'Day', 'Hour'],  
              dtype='object')
```

Generate Summary Statistics for numerical columns

```
In [26]: df.describe()
```

Out[26]:

	Timestamp	Retweets	Likes	Year	Month	Day	Hour
count	732	732.000000	732.000000	732.000000	732.000000	732.000000	732.000000
mean	2020-12-08 21:37:38.196721408	21.508197	42.901639	2020.471311	6.122951	15.497268	15.521858
min	2010-05-15 15:30:00	5.000000	10.000000	2010.000000	1.000000	1.000000	0.000000
25%	2019-03-08 17:27:30	17.750000	34.750000	2019.000000	3.000000	9.000000	13.000000
50%	2021-09-20 14:30:00	22.000000	43.000000	2021.000000	6.000000	15.000000	16.000000
75%	2023-02-26 11:48:45	25.000000	50.000000	2023.000000	9.000000	22.000000	19.000000
max	2023-10-22 20:45:00	40.000000	80.000000	2023.000000	12.000000	31.000000	23.000000
std	NaN	7.061286	14.089848	2.802285	3.411763	8.474553	4.113414

Calculate mean (Mean for Likes)

```
In [27]: df["Likes"].mean()
```

```
Out[27]: np.float64(42.90163934426229)
```

Calculate mean (Mean for all numeric values)

```
In [28]: df.mean(numeric_only=True)
```

```
Out[28]: Retweets    21.508197
Likes         42.901639
Year          2020.471311
Month          6.122951
Day           15.497268
Hour           15.521858
dtype: float64
```

Calculate median (Mean for all numeric values)

```
In [29]: df.median(numeric_only=True)
```

```
Out[29]: Retweets      22.0  
Likes        43.0  
Year         2021.0  
Month         6.0  
Day          15.0  
Hour         16.0  
dtype: float64
```

Calculate Mode

```
In [48]: df.mode()
```

Out[48]:

	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month
0	A compassionate rain, tears of empathy falling...	Positive	2018-08-22 17:20:00	Bookworm	Instagram	#Compassionate #TearsOfEmpathy	22.0	45.0	USA	2023.0	2.0
1	A playful escapade in the carnival of life, ca...	NaN	2019-04-05 17:30:00	Carnival Dreamer	NaN	#Hopeful #SeedsOfOptimism	NaN	NaN	NaN	NaN	NaN
2	Embraced by the hopeful dawn, a gardener sowin...	NaN	2020-01-05 08:45:00	Dawn Gardener	NaN	#Playful #CarnivalEscapade	NaN	NaN	NaN	NaN	NaN
3	Proudly scaling the peaks of achievement, a mo...	NaN	2021-07-01 12:10:00	Forest Dreamer	NaN	#Proud #ScalingPeaks	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	2022-07-17 06:15:00	Nature Admirer	NaN	NaN	NaN	NaN	NaN	NaN	NaN
5	NaN	NaN	NaT	Peak Conqueror	NaN	NaN	NaN	NaN	NaN	NaN	NaN
6	NaN	NaN	NaT	Rain Nurturer	NaN	NaN	NaN	NaN	NaN	NaN	NaN
7	NaN	NaN	NaT	Wind Whisperer	NaN	NaN	NaN	NaN	NaN	NaN	NaN
8	NaN	NaN	NaT	Winter Warmth	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Calculate Mode(For a particular Column)

```
In [31]: df["Sentiment"].mode()
```

```
Out[31]: 0    Positive  
         Name: Sentiment, dtype: object
```

Calculate Standard Deviation (for all numerical values)

```
In [32]: df.std(numeric_only=True)
```

```
Out[32]: Retweets      7.061286  
         Likes       14.089848  
         Year        2.802285  
         Month       3.411763  
         Day         8.474553  
         Hour        4.113414  
         dtype: float64
```

VISUALIZE DATA DISTRIBUTION

import visualization libraries

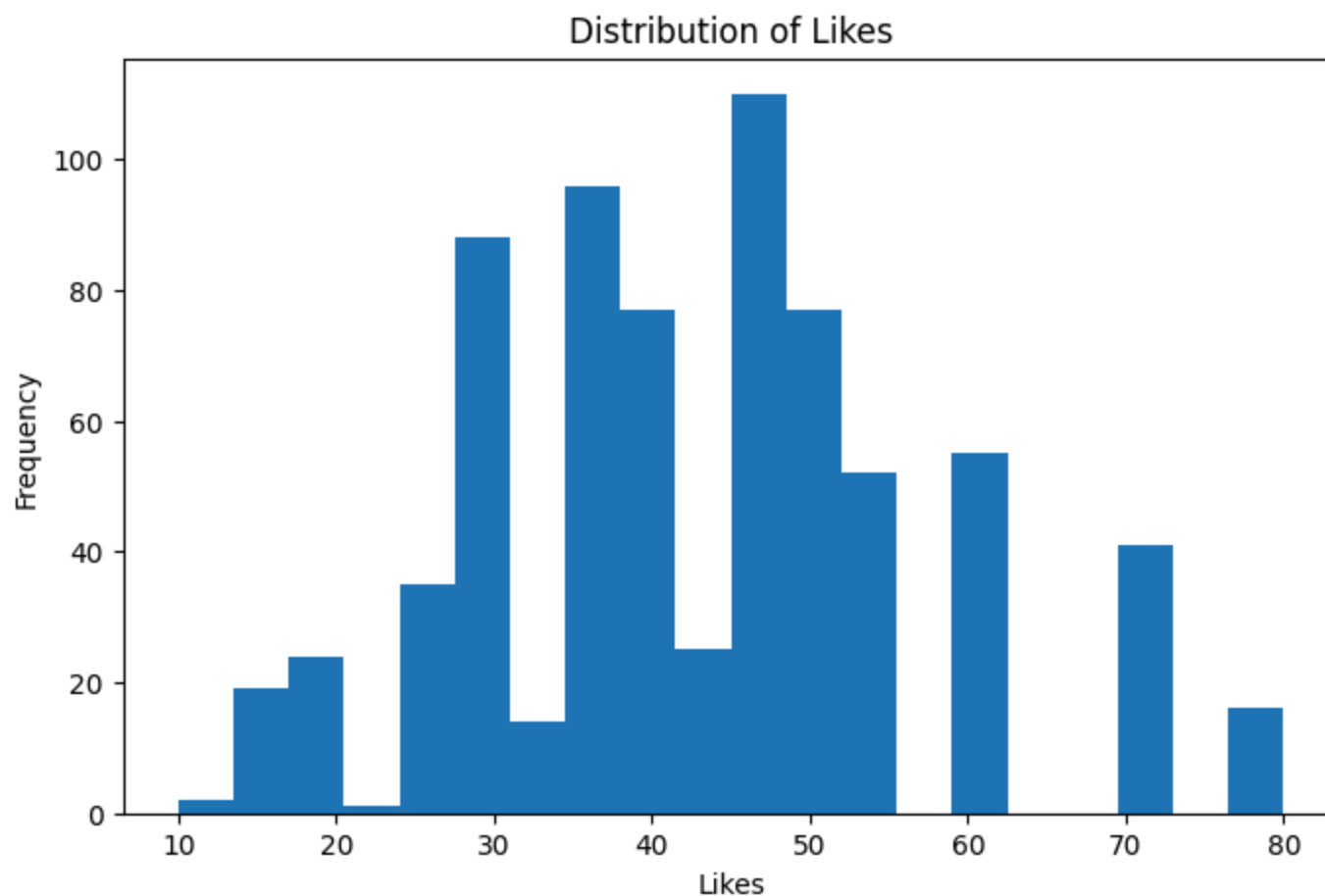
```
In [34]: import matplotlib.pyplot as plt  
         import seaborn as sns
```

```
In [35]: import matplotlib  
         import seaborn  
  
         print(matplotlib.__version__)  
         print(seaborn.__version__)
```

```
3.9.4  
0.13.2
```

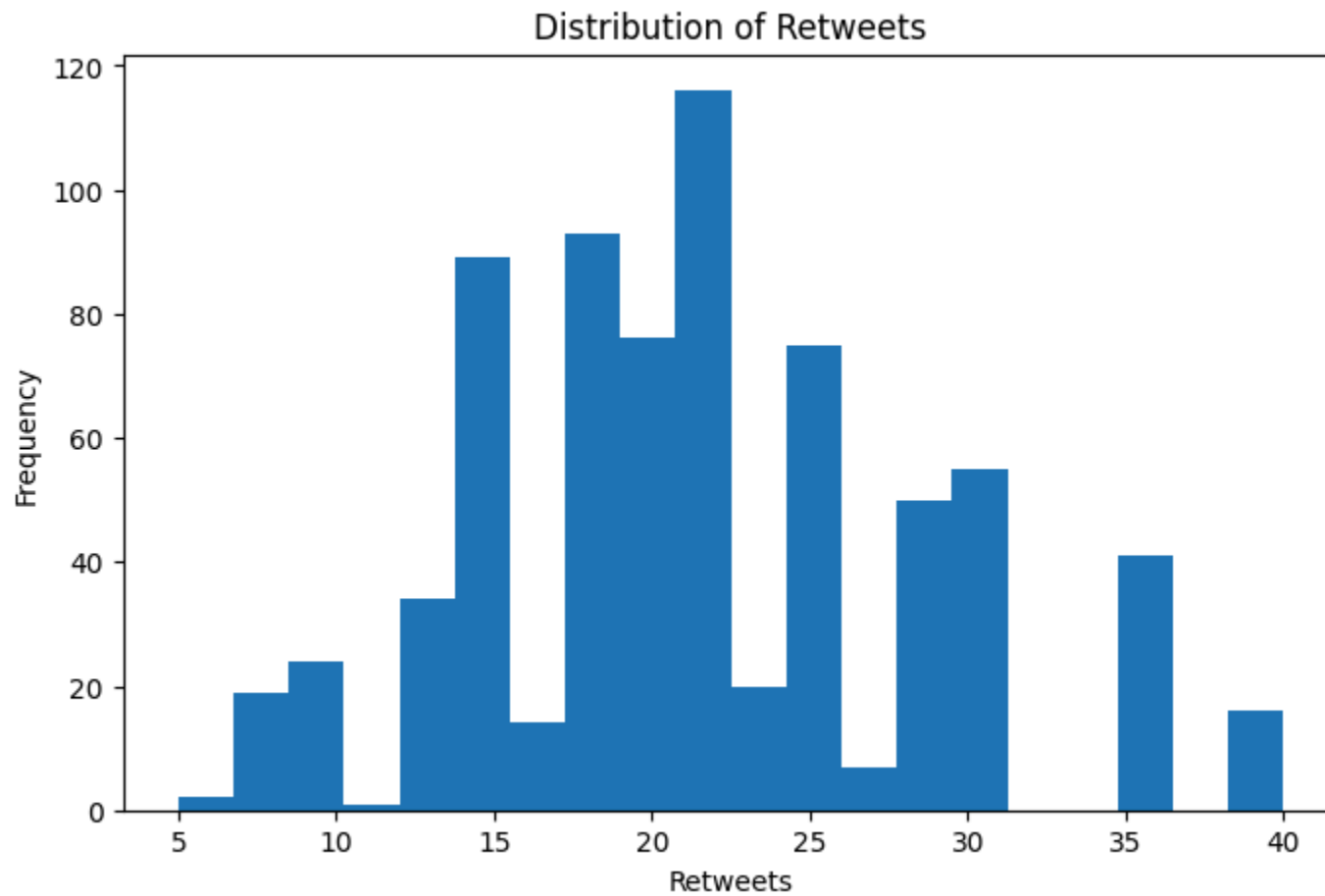

Histogram for Likes (A histogram shows how data is distributed.)

```
In [36]: plt.figure(figsize=(8,5))
plt.hist(df["Likes"], bins=20)
plt.title("Distribution of Likes")
plt.xlabel("Likes")
plt.ylabel("Frequency")
plt.savefig("likes_trend.png")
plt.show()
```



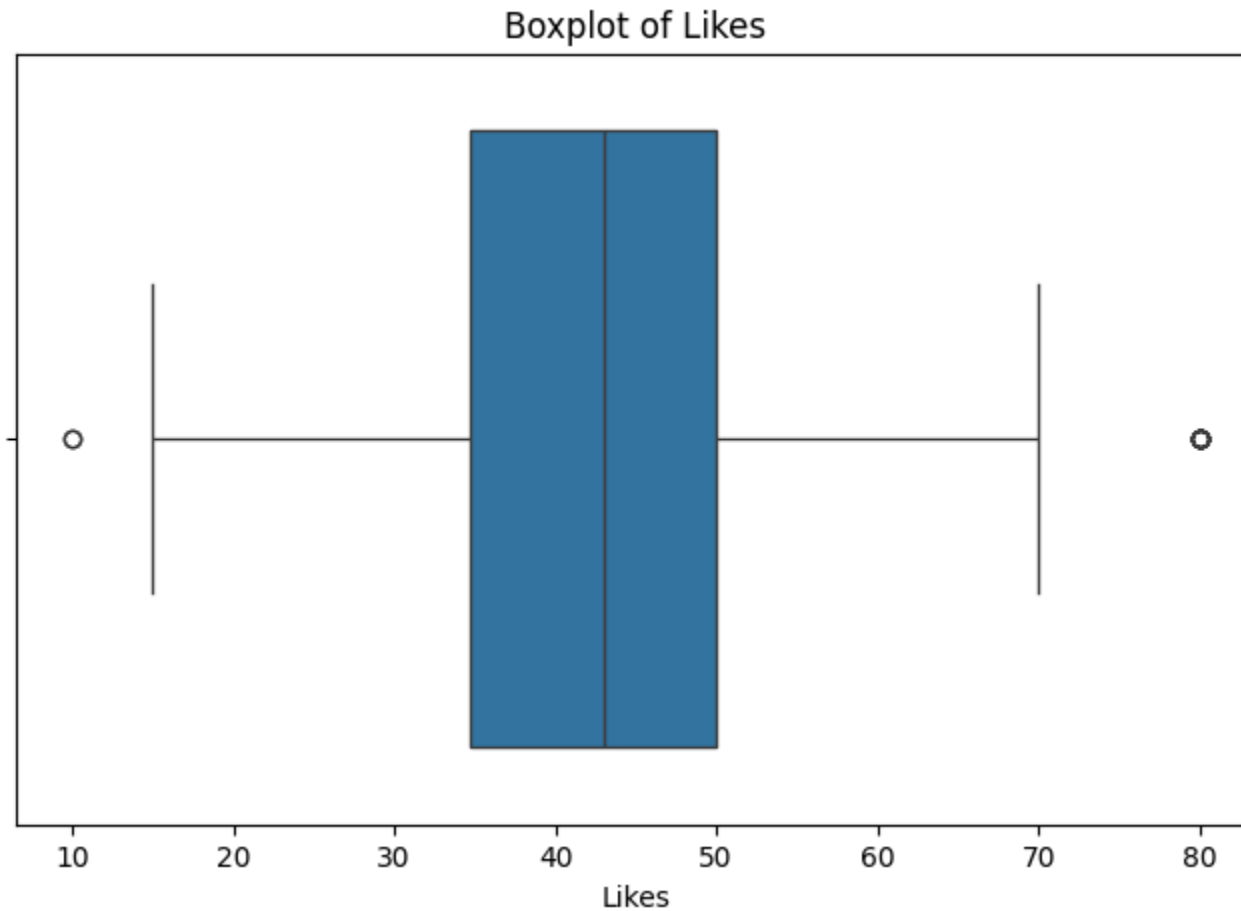
Histogram for Retweet

```
In [50]: plt.figure(figsize=(8,5))
plt.hist(df["Retweets"], bins=20)
plt.title("Distribution of Retweets")
plt.xlabel("Retweets")
plt.ylabel("Frequency")
plt.savefig("retweet_trend.png")
plt.show()
```



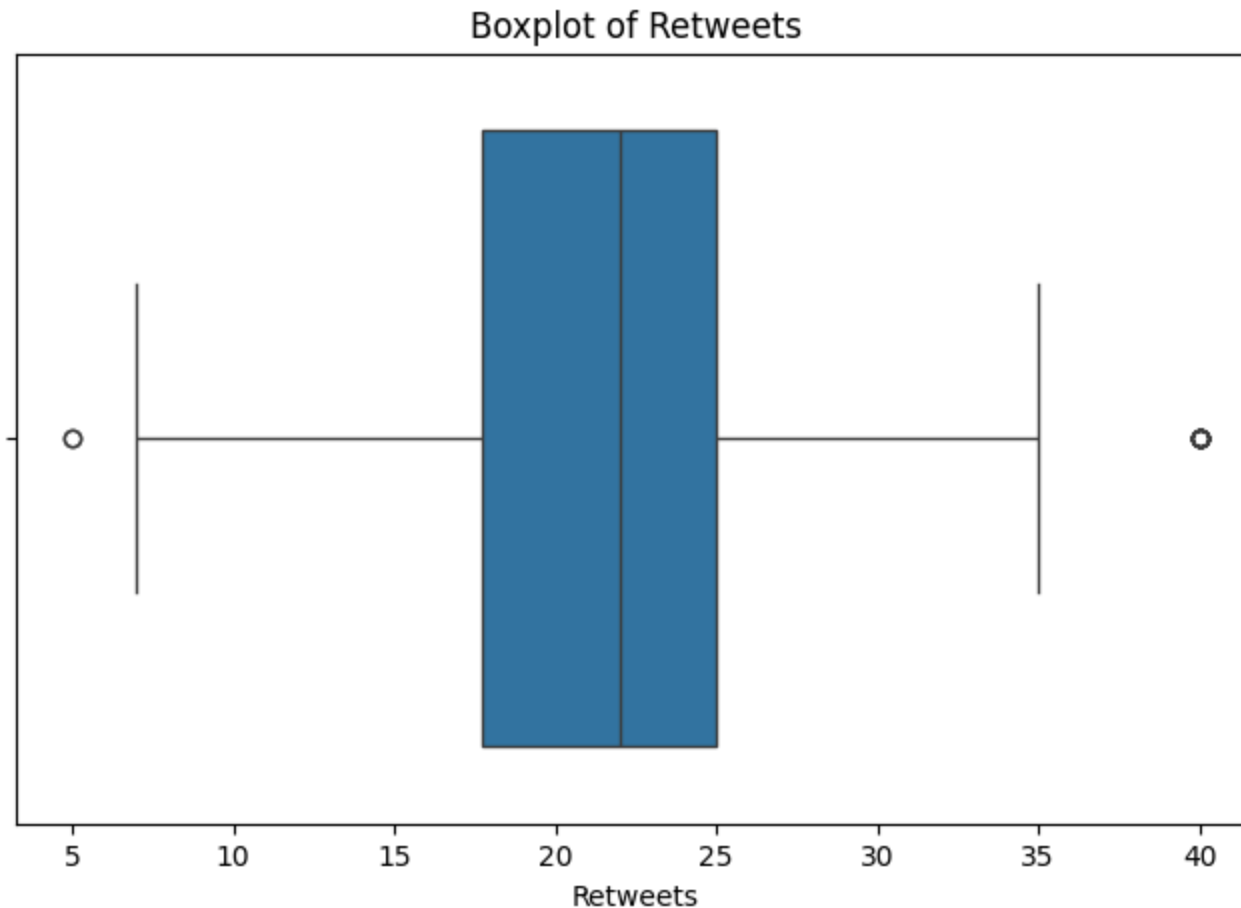
Boxplot for Likes (A boxplot identifies outliers)

```
In [52]: plt.figure(figsize=(8,5))  
sns.boxplot(x=df["Likes"])  
plt.title("Boxplot of Likes")  
plt.savefig("boxplot_likes.png")  
plt.show()
```



Boxplot for Retweet

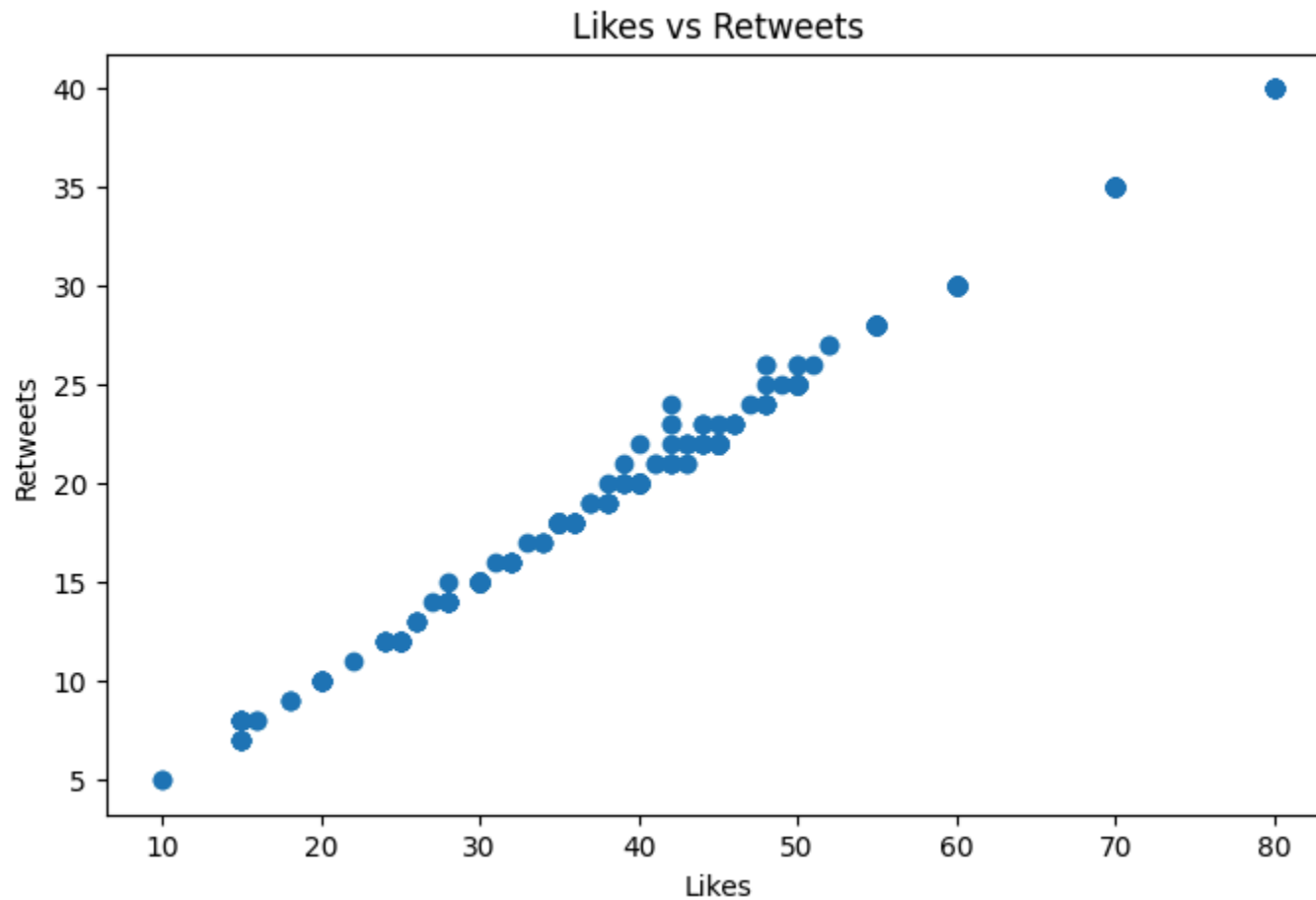
```
In [53]: plt.figure(figsize=(8,5))  
sns.boxplot(x=df["Retweets"])  
plt.title("Boxplot of Retweets")  
plt.show()
```



Scatter Plots(Scatter plots show relationships between variables)

Likes vs Retweet

```
In [54]: plt.figure(figsize=(8,5))
plt.scatter(df["Likes"], df["Retweets"])
plt.title("Likes vs Retweets")
plt.xlabel("Likes")
plt.ylabel("Retweets")
plt.savefig("boxplot_likesvsretweet.png")
plt.show()
```



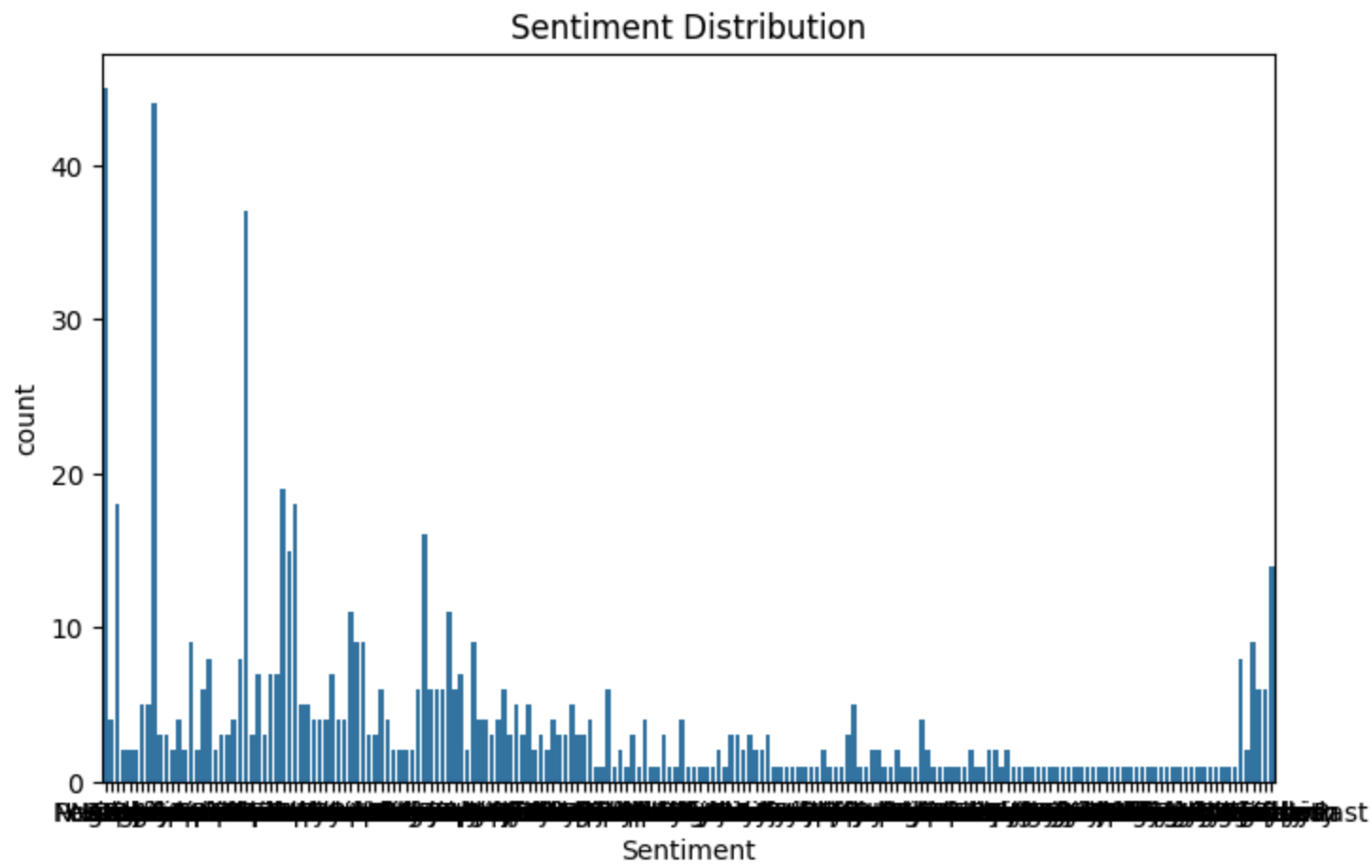
Sentiment Distribution

```
In [55]: df["Sentiment"].value_counts()
```

```
Out[55]: Sentiment
Positive          45
Joy               44
Excitement        37
Contentment       19
Neutral           18
..
Celestial Wonder   1
Nature'S Beauty    1
Thrilling Journey  1
Whispers Of The Past  1
Relief             1
Name: count, Length: 191, dtype: int64
```

Visualize

```
In [56]: plt.figure(figsize=(8,5))
sns.countplot(x="Sentiment", data=df)
plt.title("Sentiment Distribution")
plt.savefig("sentiment_distribution_count.png")
plt.show()
```



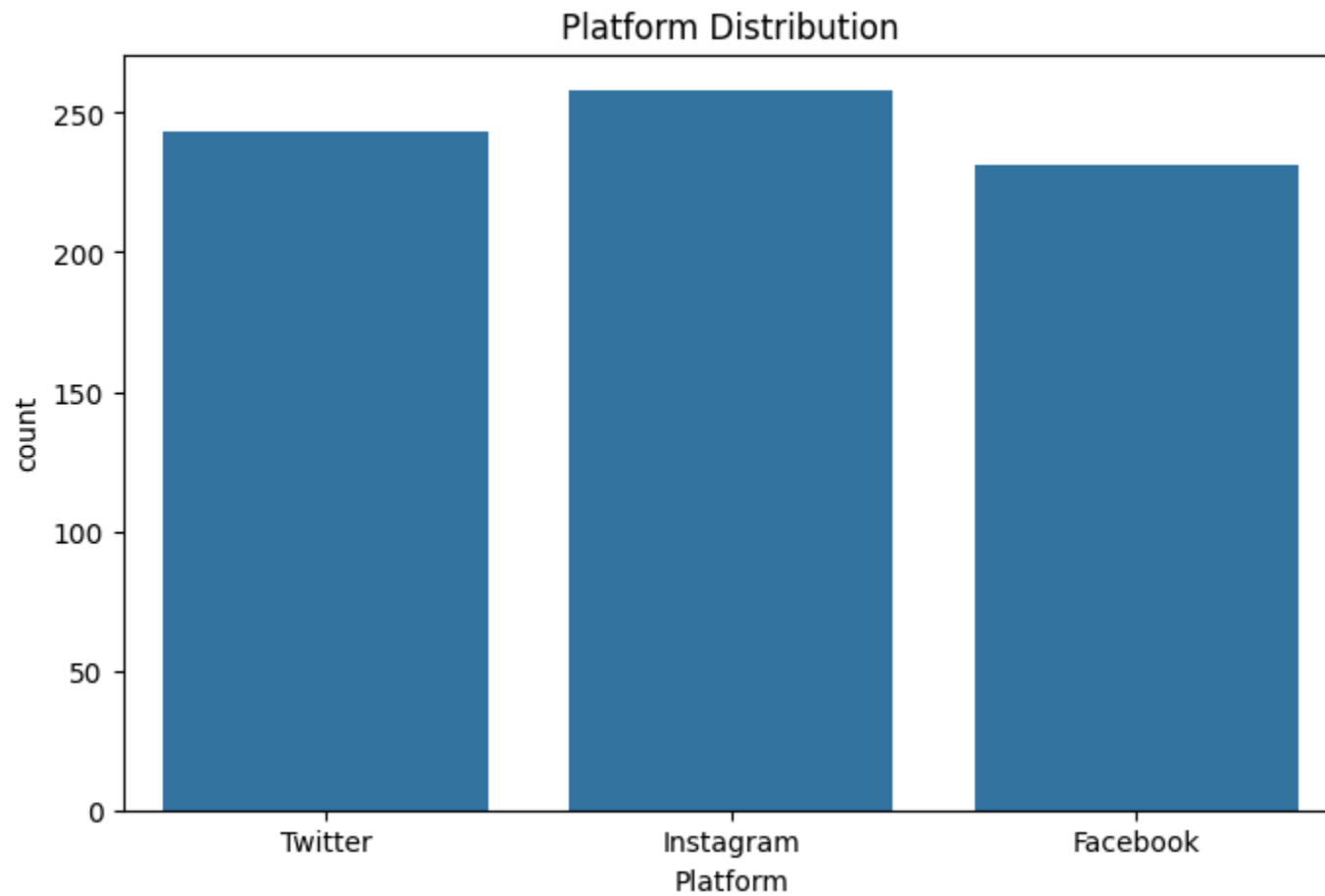
Platform Distribution

```
In [51]: df["Platform"].value_counts()
```

```
Out[51]: Platform
Instagram    258
Twitter      243
Facebook     231
Name: count, dtype: int64
```

Platform Visualization

```
In [57]: plt.figure(figsize=(8,5))  
sns.countplot(x="Platform", data=df)  
plt.title("Platform Distribution")  
plt.savefig("platform_distribution.png")  
plt.show()
```



Correlation Analysis(Find relationships among numerical variables)

Select numerical columns

```
In [59]: numerical_cols = df.select_dtypes(include=['int64','float64'])
```

View correlation matrix

```
In [60]: numerical_cols.corr()
```

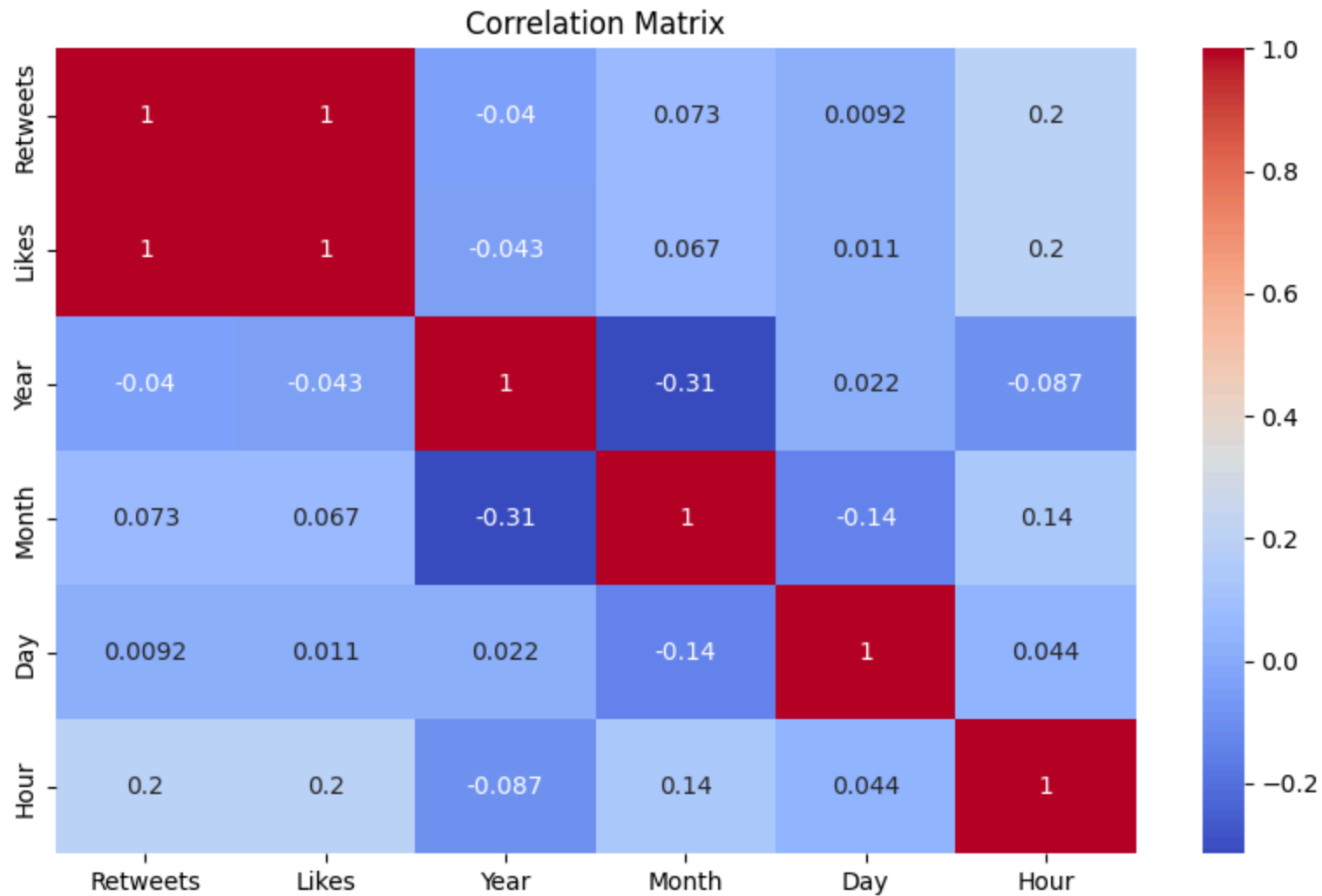
```
Out[60]:
```

	Retweets	Likes	Year	Month	Day	Hour
Retweets	1.000000	0.998482	-0.039982	0.073265	0.009213	0.196955
Likes	0.998482	1.000000	-0.043415	0.066643	0.011489	0.195331
Year	-0.039982	-0.043415	1.000000	-0.314845	0.021973	-0.087470
Month	0.073265	0.066643	-0.314845	1.000000	-0.135873	0.137835
Day	0.009213	0.011489	0.021973	-0.135873	1.000000	0.044072
Hour	0.196955	0.195331	-0.087470	0.137835	0.044072	1.000000

Correlation Heatmap

```
In [61]: plt.figure(figsize=(10,6))
sns.heatmap(
    numerical_cols.corr(),
    annot=True,
    cmap="coolwarm"
)
```

```
plt.title("Correlation Matrix")
plt.savefig("correlation_matrix.png")
plt.show()
```



STEP 3: BASIC DATA VISUALIZATION

Bar Plot *Count of Sentiment)

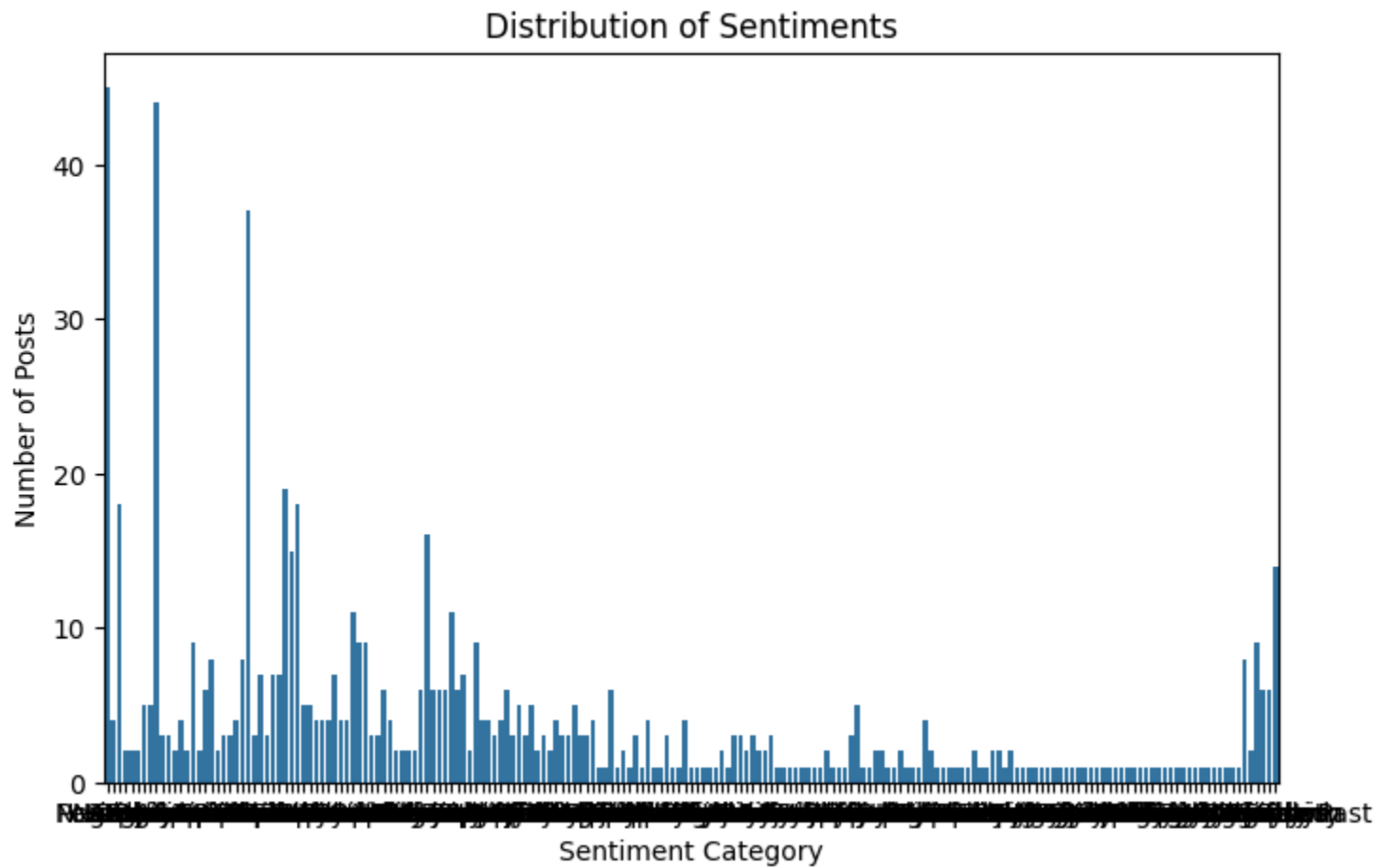
```
In [56]: plt.figure(figsize=(8,5))

sns.countplot(x='Sentiment', data=df)

plt.title("Distribution of Sentiments")
plt.xlabel("Sentiment Category")
plt.ylabel("Number of Posts")

plt.savefig("sentiment_distribution.png")

plt.show()
```



```
In [ ]: #Line Chart
```

```
In [57]: df['Timestamp'] = pd.to_datetime(df['Timestamp'])
```

```
In [ ]: #Daily Engagement Trend
```

```
In [58]: daily_likes = df.groupby(df['Timestamp'].dt.date)['Likes'].mean()
```

```
In [ ]: #plot
```

```
In [60]: plt.figure(figsize=(12,6))

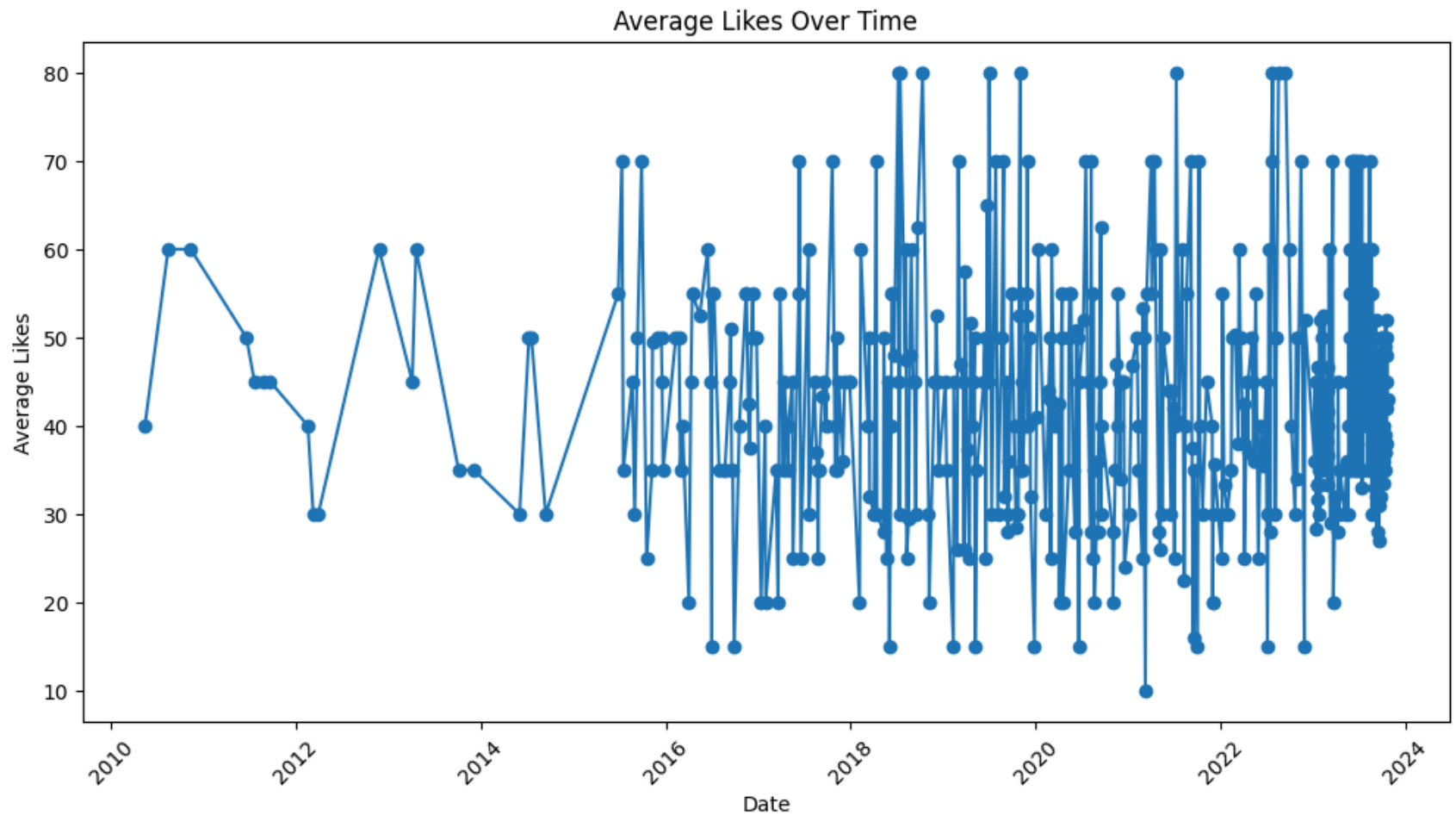
plt.plot(daily_likes.index,
         daily_likes.values,
         marker='o')

plt.title("Average Likes Over Time")
plt.xlabel("Date")
plt.ylabel("Average Likes")

plt.xticks(rotation=45)

plt.savefig("likes_trend.png")

plt.show()
```

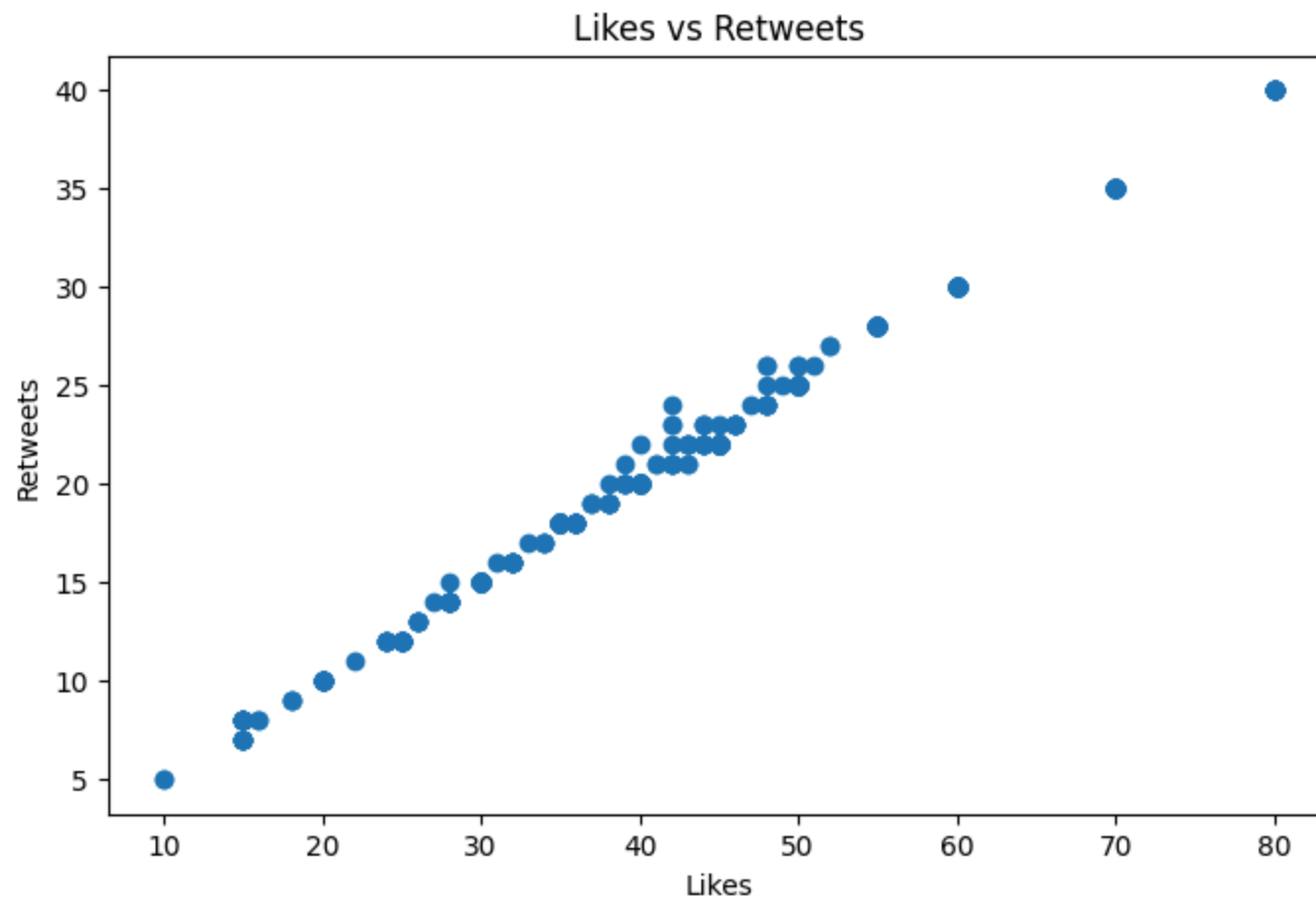


```
In [ ]: #Scatter Plot(Like and Retweet)
```

```
In [62]: plt.figure(figsize=(8,5))

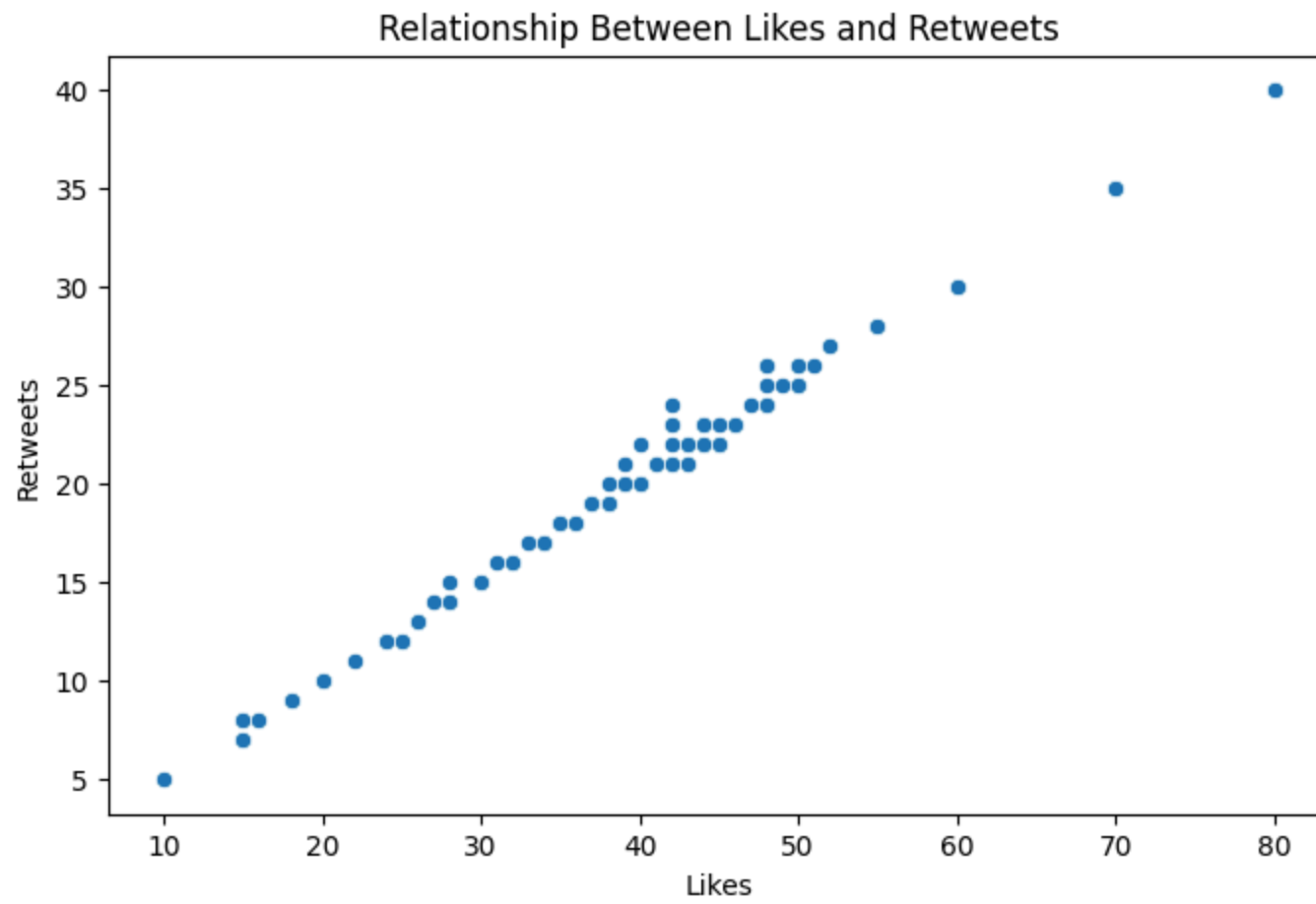
plt.scatter(df['Likes'],
            df['Retweets'])

plt.title("Likes vs Retweets")
plt.xlabel("Likes")
plt.ylabel("Retweets")
plt.savefig("scatterplot_likesvsretweet.png")
plt.show()
```



```
In [63]: plt.figure(figsize=(8,5))

sns.scatterplot(
    x='Likes',
    y='Retweets',
    data=df
)
plt.title("Relationship Between Likes and Retweets")
plt.xlabel("Likes")
plt.ylabel("Retweets")
plt.savefig("Relationship_Between_Likes_and_Retweets.png")
plt.show()
```



```
In [63]: plt.savefig("likes_retweets_relationship.png")
```

<Figure size 640x480 with 0 Axes>

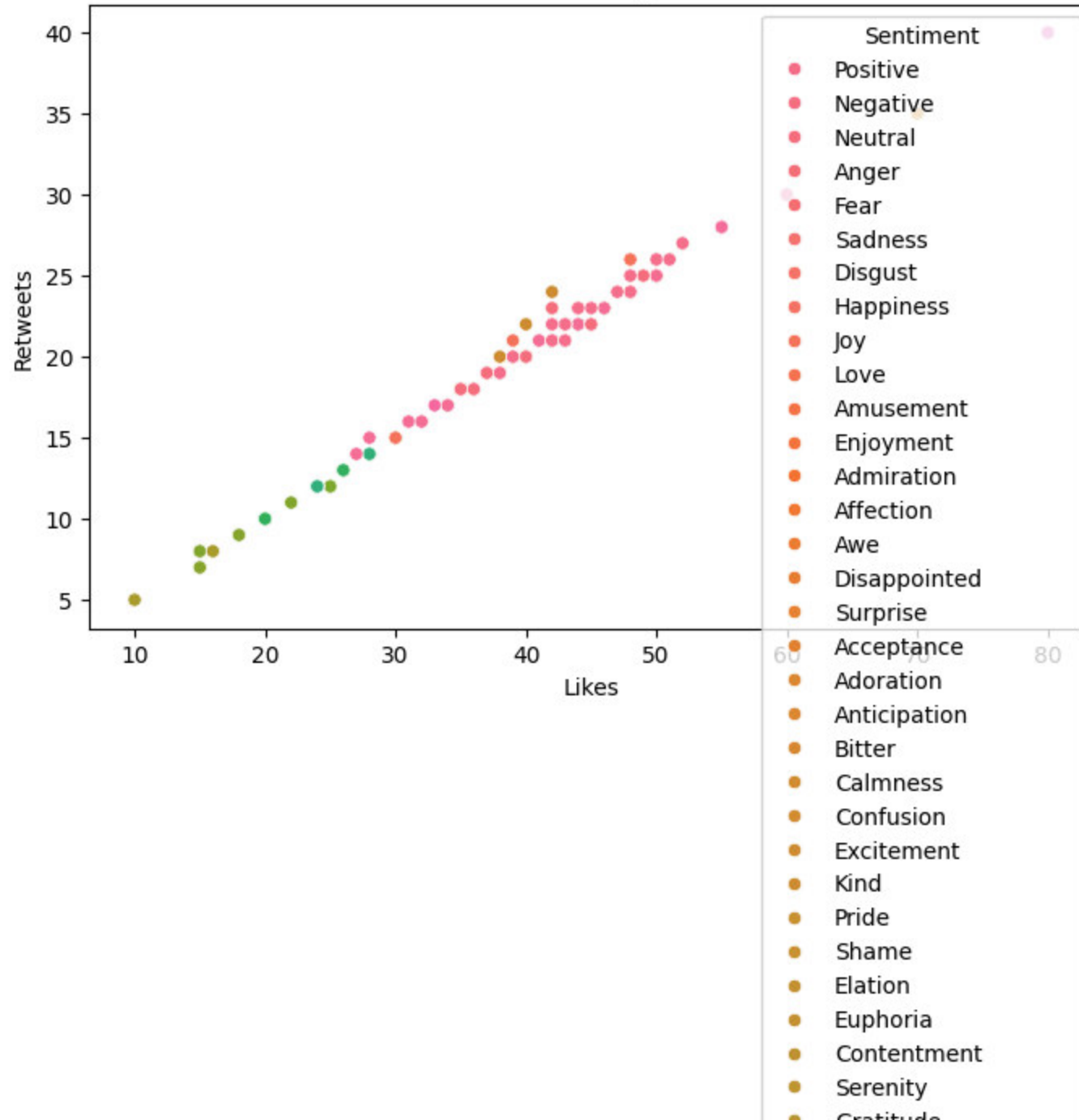
```
In [ ]: #Legend
```

```
In [64]: plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    x='Likes',  
    y='Retweets',  
    hue='Sentiment',  
    data=df
```

```
)  
plt.title("Likes vs Retweets by Sentiment")  
plt.xlabel("Likes")  
plt.ylabel("Retweets")  
  
plt.legend(title="Sentiment")  
plt.savefig("Likes_vs_Retweets_Sentiment.png")  
plt.show()
```


Likes vs Retweets by Sentiment



- Gratitude
- Hope
- Empowerment
- Compassion
- Tenderness
- Arousal
- Enthusiasm
- Fulfillment
- Reverence
- Despair
- Grief
- Loneliness
- Jealousy
- Resentment
- Frustration
- Boredom
- Anxiety
- Intimidation
- Helplessness
- Envy
- Regret
- Curiosity
- Indifference
- Numbness
- Melancholy
- Nostalgia
- Ambivalence
- Determination
- Zest
- Hopeful
- Proud
- Grateful
- Empathetic
- Compassionate
- Playful

- Free-Spirited
- Inspired
- Confident
- Bitterness
- Yearning
- Fearful
- Apprehensive
- Overwhelmed
- Jealous
- Devastated
- Frustrated
- Envious
- Dismissive
- Thrill
- Bittersweet
- Overjoyed
- Inspiration
- Motivation
- Contemplation
- Joyfulreunion
- Satisfaction
- Blessed
- Reflection
- Appreciation
- Confidence
- Accomplishment
- Wonderment
- Optimism
- Enchantment
- Intrigue
- Playfuljoy
- Mindfulness
- Dreamchaser
- Elegance

- Whimsy
- Pensive
- Harmony
- Creativity
- Radiance
- Wonder
- Rejuvenation
- Coziness
- Adventure
- Melodic
- Festivejoy
- Innerjourney
- Freedom
- Dazzle
- Adrenaline
- Artisticburst
- Culinaryodyssey
- Resilience
- Immersion
- Spark
- Marvel
- Heartbreak
- Betrayal
- Suffering
- Emotionalstorm
- Isolation
- Disappointment
- Lostlove
- Exhaustion
- Sorrow
- Darkness
- Desperation
- Ruins
- Desolation
- Loss

— Loss

- Heartache
- Solitude
- Positivity
- Kindness
- Friendship
- Success
- Exploration
- Amazement
- Romance
- Captivation
- Tranquility
- Grandeur
- Emotion
- Energy
- Celebration
- Charm
- Ecstasy
- Colorful
- Hypnotic
- Connection
- Iconic
- Journey
- Engagement
- Touched
- Suspense
- Triumph
- Heartwarming
- Obstacle
- Sympathy
- Pressure
- Renewed Effort
- Miscalculation
- Challenge
- Solace

- Breakthrough
- Joy In Baking
- Envisioning History
- Imagination
- Vibrancy
- Mesmerizing
- Culinary Adventure
- Winter Magic
- Thrilling Journey
- Nature'S Beauty
- Celestial Wonder
- Creative Inspiration
- Runway Creativity
- Ocean'S Freedom
- Whispers Of The Past
- Relief
- Embarrassed
- Mischievous
- Sad
- Hate
- Bad
- Happy

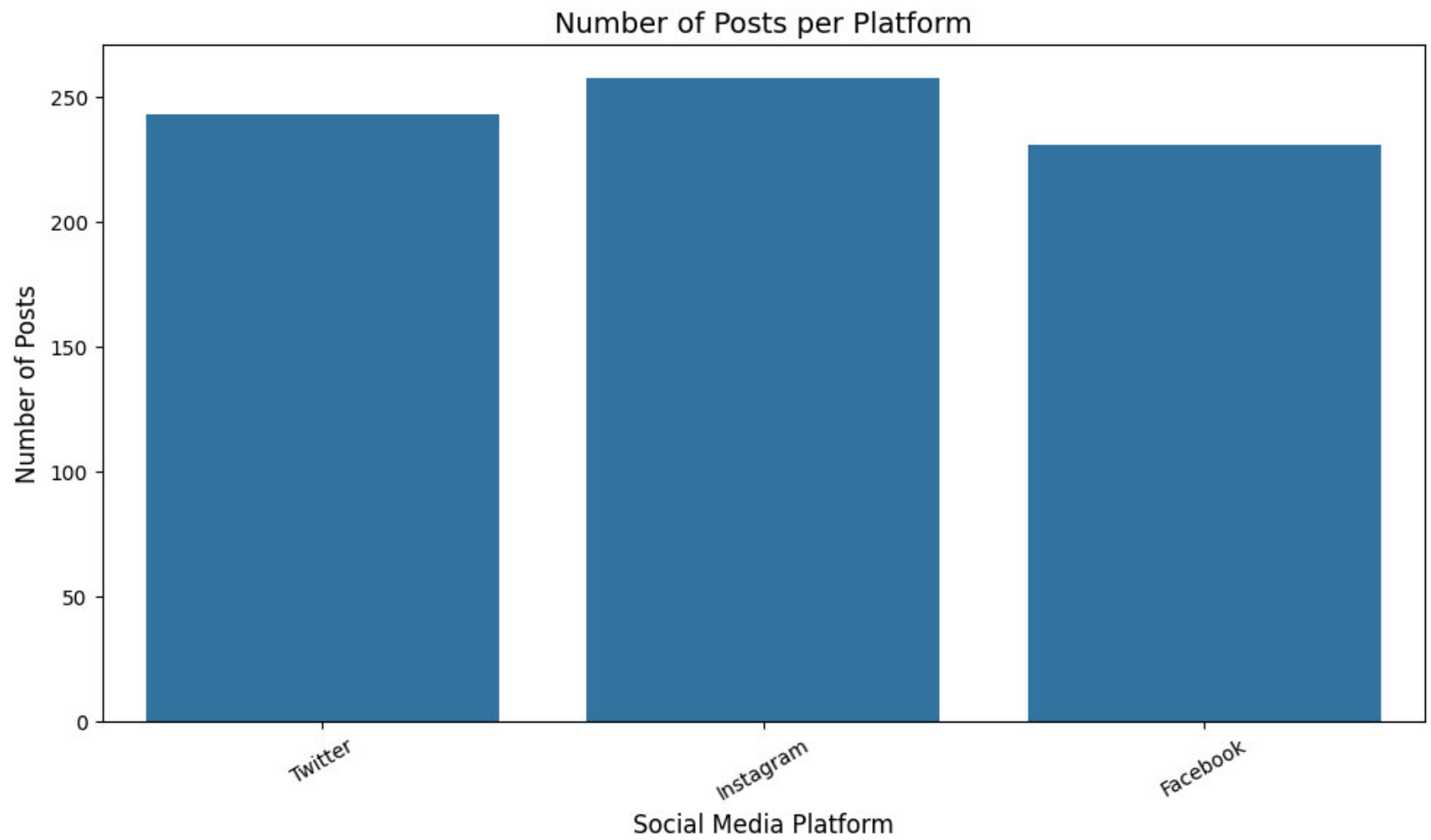
In []:

```
In [65]: plt.figure(figsize=(10,6))

sns.countplot(
    x='Platform',
    data=df
)

plt.title(
    "Number of Posts per Platform",
    fontsize=14
)
```

```
plt.xlabel(  
    "Social Media Platform",  
    fontsize=12  
)  
  
plt.ylabel(  
    "Number of Posts",  
    fontsize=12  
)  
  
plt.xticks(rotation=30)  
  
plt.tight_layout()  
plt.savefig("number_of_post_per_platform.png")  
plt.show()
```



```
In [66]: print(df.columns.tolist())
```

```
['Text', 'Sentiment', 'Timestamp', 'User', 'Platform', 'Hashtags', 'Retweets', 'Likes', 'Country', 'Year', 'Month', 'Day', 'Hour']
```

```
In [ ]:
```